Week 48 Inside and outside covers

Facilitator guide | Final student packets v3 | Prepared 3 October 2026 | Revised 23 October 2026 | Unpiloted

Mathematical overview

Finite area certificates. The sum of grid-square areas wholly inside a shape is a lower bound; the sum for a grid-square cover is an upper bound. Use aligned tiles without overlap of positive area. For the polygonal shapes here, count a cell in the cover only when it contains positive gray area. Edge-only and corner-only contact adds no cell; the boundary is already covered by neighboring closed tiles.

On nested refinement, a whole inside cell keeps all its area, while a cover cell can keep all four children or discard the outside ones. Therefore the lower bound cannot fall and the upper bound cannot rise. Their gap is an explicit uncertainty certificate. Equality of limiting inner and outer contents requires a measurability condition; finite refinement alone does not prove it for every set.

What the markings guarantee. If w cells are whole and $p \geq 1$ partial, the area lies strictly between w and w + p, and every area in between is reached by some shape with the same marked cells. A split removes at most its own cell's one unit of uncertainty: all of it only when each child is whole or outside, possibly none. So refining need not halve the gap: the triangle's and diamond's gaps halve, the trapezoid's falls from 12 to 3.

Keep the unit fixed. One big square has area 1; four half-side squares have total area 1. The main boards are 120 mm square, with 30 mm big tiles and 15 mm small tiles; the same-markings drawing boards are 85 mm. A small tile has area 1/4, not 1/2. All numerical bounds in the older packets use big-square units.

Discovery by band. K-1 classifies cells, selects useful refinements and changes area without changing a coarse certificate. Grades 2-3 tightens bounds, finds that refining need not halve the gap, and defeats claims of exact area. Grades 4-5 optimizes a limited refinement budget and explains monotonicity. Counting suggests a pattern; cell containment proves the bounds.

Prepare and launch

Per pair: one tile sheet, week-48-materials.pdf, printed at actual size on card stock and cut apart: 16 big 30 mm and 48 small 15 mm squares. The trapezoid's cover takes all 16 big tiles and K-1 Problem 3's small-square cover of the triangle takes 36 small (no fine cover here takes more than 40), so each pair needs a set. Per table: chosen student pages, pencils or two colors, transparent sleeves or tracing paper. Children may tile a fine cover or shade it on the board. Print at actual size; check the sheet's 10 cm line.

Readiness. Start with visual whole/partial/outside decisions and adult counting. Four-small-to-one-big regrouping is enough for older bounds; fraction symbols are optional. Keep one adult at each of the current three tables.

Launch, 4 minutes, after the run. Let children cover a few cells. Use the three printed sample cells to show whole gray, partly gray and outside. Demonstrate an edge touch that needs no extra tile, then reassemble four small squares into one big one. Ask, "What can your tiles prove about the amount of gray?"

Shared triangle and diamond keys

Final task	Whole inside cells	Cover cells	Bounds in big units
All bands P1 triangle	6 big	10 big	6 ≤ area ≤ 10
Older P2 diamond	4 big	12 big	4 ≤ area ≤ 12
All bands P3 triangle	28 small	36 small	7 ≤ area ≤ 9
Older P4 diamond	24 small	40 small	6 ≤ area ≤ 10

Problem 1. For the coarse triangle, successive rows from top to bottom contain 0,1,2,3 whole cells and 1,2,3,4 cover cells. Four diagonal cells are partial. **Older Problem 2:** the diamond has four central whole cells and eight partial cells. Therefore its area is certainly more than 3 and less than 13; an exact-area formula is unnecessary.

Problem 3. Each of the four partial big triangle cells becomes one whole, two partial and one outside small cell. This adds one big unit to the inside bound and removes one from the cover. **Older Problem 4:** each of the eight partial big diamond cells has the same four-child pattern. Its uncertainty gap falls from 8 to 4 big units. The triangle’s gap falls from 4 to 2.

K-1 specific keys

Problem 2. The trapezoid’s coarse cover uses all 16 big cells. Refine **any two of the four corner cells**: each has one partial child and three outside children, so remove **6 small tiles** in total. No covered big cell can lose all four children, so three removable children per chosen tile is the absolute maximum. The remaining cover has area 16 - 6/4 = 14.5 big units; children may simply count the removed pieces. The bottom corner cells hold only a gray sliver at most 2.5 mm wide beside the slanted side: they are partial, so their big tiles stay in the cover, and after a split one small tile stays over the sliver.

Problem 4. Each of the diamond’s eight partial big cells has one wholly outside small child. Choose four such children in different big cells and bulge the nearby diagonal boundary slightly into each, leaving part of each chosen small cell unshaded. Keep all changes within those four partial big cells. This makes exactly four previously outside small cells partial while every big-cell status stays unchanged. Up to eight could change this way, one in each partial big cell. There is no unique changed shape.

Problem 5. Each shape has its own board, so both stay visible; Problem 6 continues on the larger shape’s board. In the same three columns, shade a rectangle one full row high plus a shallow part of the next row. Width 3 and height 7/6 gives area **3.5**; width 3 and height 11/6 gives **5.5**. Both have exactly three whole and the same three partial big cells. Folding or drawing a visibly thin versus nearly full extra row suffices; children need not measure sixths.

Problem 6. Yes. Raise the 11/6-height rectangle to height 23/12, for example: area 5.75. Its three partial cells remain partial. One can approach area 6 as closely as desired while leaving a positive unshaded strip, but cannot attain 6 while all three remain genuinely partial. Thus 6 is a **supremum, not a largest allowed area**.

Older specific keys

Grades 2-3 Problem 5. On the big grid the trapezoid has 4 whole and 12 partial cells: bounds 4 and 16. On the small grid it has 24 whole and 12 partial small cells: bounds 6 and 9 big units. The gap falls from 12 to 3, to a quarter, where the triangle's and diamond's halved: the top and bottom middle cells lose all their uncertainty and the corners three quarters of it (table under grades 4-5 Problem 5). Each bottom corner cell holds a gray sliver at most 2.5 mm wide; it is partial, so it is in both covers. Splitting never loosens a bound, but how much it helps depends on the shape.

Grades 2-3 Problem 6. Width-4 rectangles with heights 9/8 and 15/8 have areas **4.5 and 7.5**. Both use one whole row and part of the next: exactly four whole and four partial big cells, in the same positions.

Grades 2-3 Problem 7. Neither less than 4 nor more than 8 is possible. The coarse certificate is $4 \le \text{area} \le 8$. For ordinary drawings with positive gray and positive white area in each partial cell, actually **$4 < \text{area} < 8$**. These are strict because some gray must be added beyond the four whole cells and some white must remain below the eight-cell cover. Areas can approach either endpoint; neither is attained with four genuinely partial cells.

Grades 4-5 Problem 5. Number columns 1-4 from left to right and rows 1-4 from top to bottom. Refine any three of **(2,1), (3,1), (2,4), (3,4)**. Each of these top/bottom middle cells splits into two whole and two outside small cells, eliminating its entire one-unit uncertainty. The initial bounds 4 and 16 become **5.5 and 14.5**, so the gap falls from 12 to **9**. Check that children count the bottom corner cells as partial: the gray there is a sliver at most 2.5 mm wide.

Trapezoid cell type	Small children I / P / O	Gap reduction per big cell
Top/bottom middle, 4 cells	2 / 0 / 2	1
Four corner cells	0 / 1 / 3	3/4
Left/right middle, 4 cells	0 / 2 / 2	1/2
Central whole, 4 cells	4 / 0 / 0	0

Here I/P/O means inside/partial/outside. A big cell can remove at most its entire one-unit gap, so three refinements cannot gain more than 3. The displayed choice achieves 3 and proves optimality. Notice that this objective differs from K-1's removal-only objective, where corners are best.

Older specific keys, continued

Grades 4-5 Problem 6. No to both questions. Every old inside square is exactly its four inside children. New inside children may add area, never remove old inside area. Every retained cover child lies in its parent cover square. Deleting outside children can only reduce the cover. This assumes the same shape and a true subdivision of the same grid.

Grades 4-5 Problem 7. The trapezoid is a 2-by-3 rectangle and two right triangles with legs 1/2 and 3, each of area 3/4; the two triangles together make a 1/2-by-3 rectangle. Its exact area is **15/2 = 7.5 big-square units**, between the first bounds 4 and 16 and between the bounds after three splits, 5.5 and 14.5. Refinement improves the certificate, not the shape's actual area.

Optional hints and sources

If stuck, ask "Which uncertain tile would tell us the most if split?" Keep inside and cover colors separate. For shape-changing tasks, preserve the same labeled cells and try thin strips. These are optional hints, not a mandatory method.

A. Sapozhnikov, *Mathematics 3*, §1.2, pp. 2-4, develops inner/outer Jordan content and nested dyadic boxes: author lecture notes. Our finite boundary-touch convention and all polygon counts, optimal refinements and rectangle witnesses were checked independently by exact rational clipping. All material handling and pacing are unpiloted proposals; no physical testing is claimed.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First route: K-1 P1, then P3 to compare coarse and fine triangle covers; grades 2-3 and 4-5 P1-2, then one fine comparison in P3-4. On return, K-1 can choose a limited refinement (P2), alter a boundary (P4), or vary area with the same coarse markings (P5-6). Grades 2-3 can test on the trapezoid whether refining always halves the gap (P5) and build counterexamples to exact-area claims (P6-7). Grades 4-5 can optimize the refinement budget (P5), justify monotonicity (P6) and find the trapezoid's exact area between its bounds (P7). Four small squares retain one big-square area unit.

A first-visit hour

0-5 running around; 5-9 launch with the tiles and the three sample cells; 9-27 coarse covers; 27-47 refine or change a shape; 47-55 compare certificates and counterexamples; 55-60 reset. One sound before-and-after cover is enough for a useful session.